Microeconomics

Lecture 2 · The consumer problem

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From preferences to an actual choice

- Budget sets, existence and Marshallian demand.

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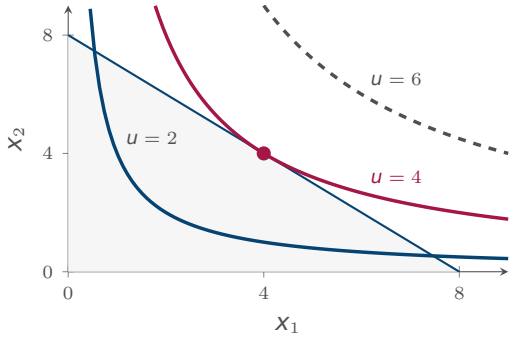
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From preferences to an actual choice

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- The Lagrangian, saddle points and Kuhn–Tucker conditions, with two proofs.
- Budget exhaustion and worked examples; then indirect utility, the envelope theorem and Roy’s identity.

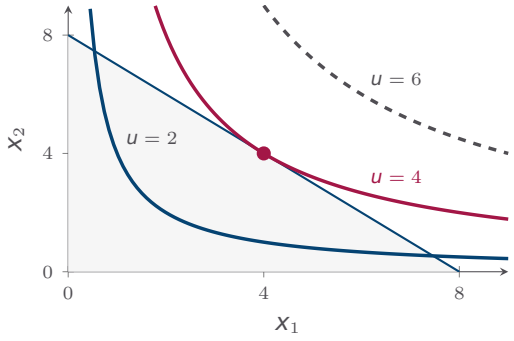
The picture we already know



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$p = (1, 1)$, $m = 8$.

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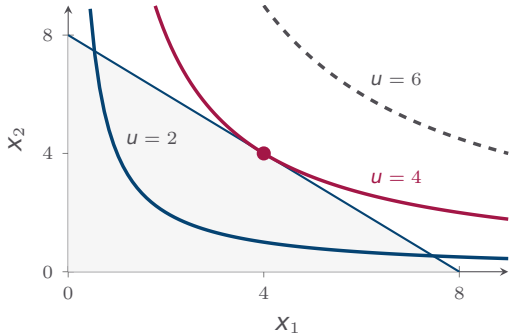


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Best bundle: $(4, 4)$.

Utility is 4.

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Here $p \gg 0$ and $m > 0$. The feasible set is

$$B(p, m) = \{x \geq 0 : p \cdot x \leq m\}.$$

What did we assume when we wrote that?

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What changes for a subscription, a quantity discount or an indivisible purchase?

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The same bundles are affordable. Only the currency units have changed.

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A closed and bounded subset of finite-dimensional Euclidean space is compact.

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Existence does not yet imply uniqueness, an interior solution or differentiability of demand.

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Can you give a case with multiple maximisers even though the budget is compact?

For $u = x_1 + x_2$, $p = (1, 1)$ and $m = 1$, every bundle on $x_1 + x_2 = 1$ is optimal.

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It is the earlier choice rule applied specifically to budget menus: $D(p, m) = C(B(p, m))$.

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Example: $(p, m) = ((2, 1), 12)$ and $((200, 100), 1200)$.

Let a multiplier enforce the budget

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One variable chooses consumption; the other enforces the budget. Their directions of optimisation differ.

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This identity does not authorise exchanging max and inf.

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Left inequality: x^* maximises ℓ over the consumption domain, holding λ^* fixed.

Right inequality: λ^* minimises ℓ over nonnegative multipliers, holding x^* fixed.

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Therefore $\lambda^*(m - p \cdot x^*) = 0$ and $\ell(x^*, \lambda^*) = u(x^*)$.

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This direction needs no concavity or differentiability: the saddle inequalities are already global.

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We now make the bounds $x_i \geq 0$ explicit using multipliers μ_i. → KKT back to a saddle

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If both goods are consumed in a smooth optimum, these marginal returns must agree.

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For the smooth argument, assume u is differentiable on an open neighbourhood of the candidate.

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Stationarity is only one group of conditions. We must also check signs, feasibility and slackness.

The four Kuhn–Tucker groups

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Complementary slackness:

$$\lambda^*(m - p \cdot x^*) = 0, \qquad \mu_i^* x_i^* = 0.$$

Complementary slackness: read the products

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Does a binding constraint always have a strictly positive multiplier?

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This is an interior condition, not a universal rule for every optimum.

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Why would a rule that sets every marginal return equal miss some solutions?

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Quasiconcavity alone does not make every stationary point optimal. → Sufficiency proof → Counterexample

Theorem 1: KKT is necessary

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We prove this directly: construct the multipliers, prove their signs, then verify complementary slackness.

Construct the budget multiplier from an optimum

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Feasible spending transfers between two consumed goods show that $u_i(x^*)/p_i = \lambda^*$ whenever $x_i^* > 0$.

Also $\lambda^* \geq 0$: otherwise $u_j(x^*) < 0$, and consuming slightly less of good j would be affordable and strictly improve utility.

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The reverse strict inequality is ruled out by the reverse transfer. Thus $u_i(x^*)/p_i = u_j(x^*)/p_j = \lambda^*$.

Construct the quantity-bound multipliers

For each good, define

$$\mu_i^* := \begin{cases} 0, & x_i^* > 0, \\ \lambda^* p_i - u_i(x^*), & x_i^* = 0. \end{cases}$$

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The remaining question: why are these constructed μ_i^* nonnegative? We have not assumed their sign.

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The unconsumed good would give more marginal utility per unit of spending.

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$$\frac{d}{d\varepsilon} u(x(\varepsilon))\Big|_{0+} = \frac{u_i(x^*)}{p_i} - \frac{u_j(x^*)}{p_j} = -\frac{\mu_i^*}{p_i} > 0.$$

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A sufficiently small transfer strictly improves utility: contradiction. Thus $\mu_i^* \geq 0$, with equality allowed.

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For every i, the construction gives $\mu_i^* x_i^* = 0$: either the quantity or its multiplier is zero.

Feasibility came from x^*; stationarity, multiplier signs and slackness have now all been proved.

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This completes the necessity proof, including all corners and the zero bundle.

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Let u be concave on $\mathbb{R}^L_+$ and differentiable on a neighbourhood of a candidate x^*.

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The global argument uses concavity of the chosen index, not merely convexity of preferences.

Fix the multipliers and look only at quantities

Define $f(x) = \mathcal{L}(x, \lambda^*, \mu^*)$, holding the multipliers fixed.

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The constructed stationarity conditions give $\nabla f(x^*) = 0$. Why is that a global maximum?

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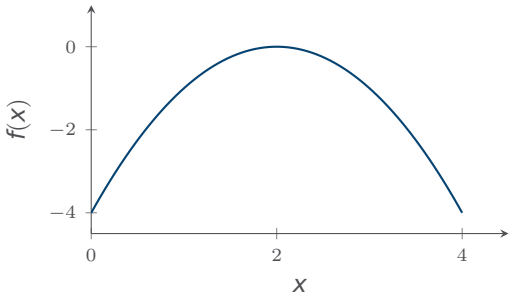
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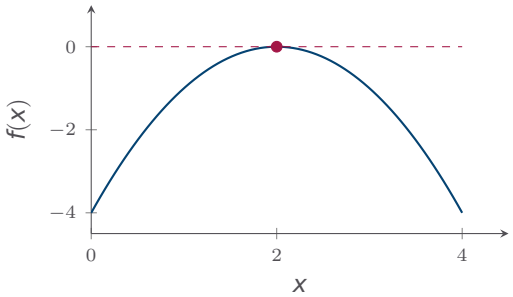
Thus x^* globally maximises $\mathcal{L}(\cdot, \lambda^*, \mu^*)$ on that domain. Uniqueness is not asserted.

A horizontal tangent above the whole graph



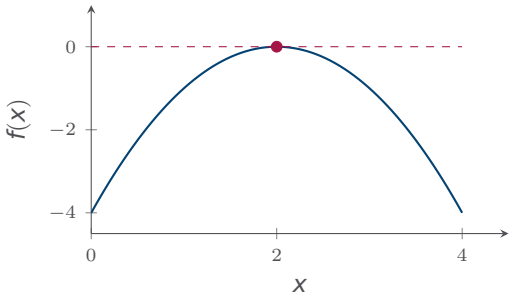
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The horizontal tangent is an upper bound on the entire graph.
Zero gradient means a flat tangent, not a constant function nearby.

Proof 2: return from the Lagrangian to utility

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The same slackness and feasibility minimise $\mathcal{L}(x^*, \lambda, \mu)$ over nonnegative multipliers: the saddle interpretation is recovered.

Budget exhaustion: Walras' law

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Assume an optimum exists. What would happen if it left strictly positive budget slack?

The key: with slack, every sufficiently small change in the consumption domain remains affordable.

Proof: LNS and budget exhaustion

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LNS supplies a strictly better such y in the consumption domain, contradicting optimality.

→ Back to budget exhaustion

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For every affordable w, optimality gives $x \succeq w$ and $y \succeq w$.

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Convex preferences imply $z_t \succeq w$ for every such w. Thus $z_t \in D(p, m)$.

Hence $D(p, m)$ is convex. It can still be a whole line segment.

Strict convexity: at most one optimum

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Does strict convexity alone establish existence?

No. Continuity and compactness gave existence; strict convexity gives at most one optimum.

Proof: uniqueness from strict convexity

Suppose distinct $x, y \in D(p, m)$. Their midpoint z is feasible because the budget is convex.

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This contradicts x being optimal. There is at most one optimum; existence gives a unique one.

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Smooth interior problem: marginal conditions can identify a candidate.

Always check feasibility, boundary cases and a reason why the candidate is globally optimal.

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What can go wrong if we check only the interior first-order equations?

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An axis bundle gives zero utility; a strictly positive affordable bundle gives positive utility.

Every optimum is therefore positive. We may now take logs and use an interior method.

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The log representation is concave on the positive domain.

The bundle does not change, but the numerical utility and multiplier generally do.

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$$x_1(p, m) = \frac{\alpha m}{p_1}, \qquad x_2(p, m) = \frac{(1-\alpha)m}{p_2}.$$

Interpret the solution

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Expenditure shares are constant in this specification.

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Which of these predictions is a modelling restriction rather than an accounting identity?

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Does an unchanged quantity of good 2 mean that its utility contribution is unchanged?

Example 2: perfect substitutes

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If the ratios are equal, every budget-line bundle is optimal.

Verify a corner with the full KKT system

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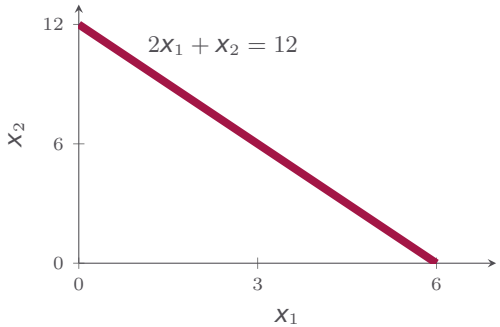
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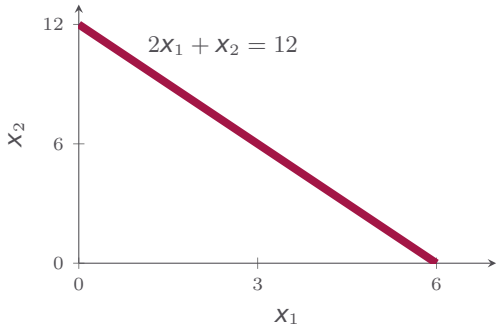
The zero quantity is optimal precisely because its marginal return is too low.

Equal returns: a whole set of demand



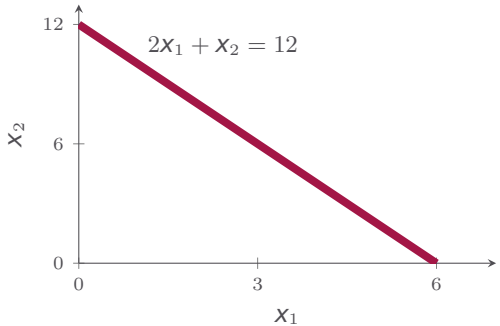
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Every budget-line bundle gives utility 12.
$D(p, m)$ is a segment. Every optimal bundle gives utility 12.

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Budget exhaustion gives $a = m - rq$ and $0 \le q \le m/r$.

Solve first, then respect the quantity bounds

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At $r = 1$: income 2 gives $(a, q) = (0, 2)$; income 8 gives $(5, 3)$.

At $m = 4 - r$, both formulas agree; a binding bound may have zero multiplier.

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The highest affordable target is $t = m/(p_1 + p_2)$.

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The optimum is a kink: the target-cost argument replaces a smooth tangency calculation.

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5. **Interpret:** quantities, expenditure shares and reactions to parameters.

From a classroom solution to a structural model

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Fitting one choice does not establish the functional form. Use more variation and test the restrictions.

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It is a function of the environment, not of a chosen bundle.

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- More wealth weakly increases v; under LNS the increase is strict.
- v is continuous on the positive-price, positive-wealth domain.

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Maximising the same utility over a smaller set cannot give a larger value: $v(q, m) \leq v(p, m)$.

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So y is affordable at the new wealth and $v(p, m') \geq u(y) > v(p, m)$.

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With $g(p, m, z) = u((m/p_1) z_1, \ldots, (m/p_L) z_L)$, we have $v(p, m) = \max_{z \in \Delta} g(p, m, z)$.

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No differentiability or uniqueness of the optimiser was needed.

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So the mixed budget cannot deliver utility above both endpoint values. → Proof

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Its utility cannot exceed $\max\{v(p, m), v(q, n)\}$. Maximise over the mixed budget.

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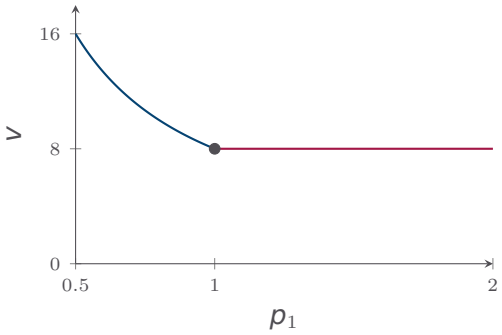
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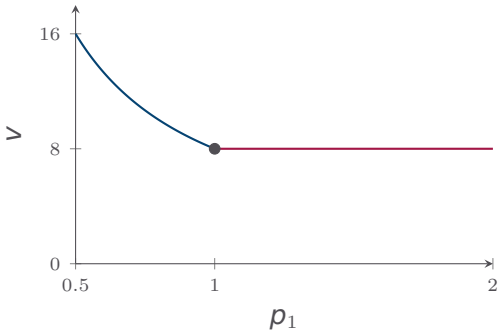
→ More than two goods

A continuous value need not be differentiable



$u = x_1 + x_2$, $p_2 = 1$, $m = 8$.

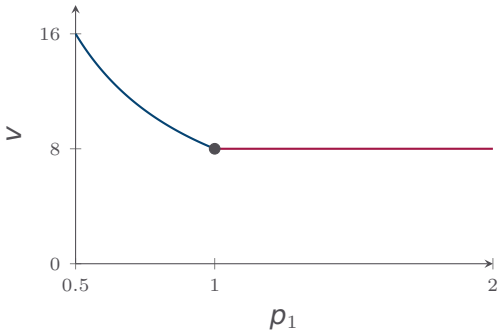
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At $p_1 = 1$, the consumer is indifferent along the budget line. The value has a kink.

The envelope question

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First a theorem for a general constrained problem; then its consumer application.

A general parameterised optimisation problem

$$V(\theta) = \max_x f(x, \theta) \quad \text{subject to} \quad g_j(x, \theta) \geq 0, \quad h_k(x, \theta) = 0.$$

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Use $\lambda_j \geq 0$ for inequalities and unrestricted ν_k for equalities.

The general Lagrangian

$$\mathcal{L} = f(x, \theta) + \sum_j \lambda_j g_j(x, \theta) + \sum_k \nu_k h_k(x, \theta).$$

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This is a sufficient smooth-branch regularity assumption, to be checked in an application.

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On the right, hold decisions and multipliers fixed when taking the partial derivative.

Thus $V_{\theta_\ell} = f_{\theta_\ell} + \sum_j \lambda_j^* (g_j)_{\theta_\ell} + \sum_k \nu_k^* (h_k)_{\theta_\ell}$, evaluated at the optimum.

Proof: differentiate along an optimal branch

Feasibility and slackness give $V(\theta) = \mathcal{L}(x^*(\theta), \lambda^*(\theta), \nu^*(\theta); \theta)$.

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Stationarity kills the $\mathcal{L}_x$ term; equality feasibility kills the h_k terms.

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Every extra term is zero. Hence

$$\frac{dV}{dt} = \mathcal{L}_t.$$

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Wealth relaxes the budget; a higher price tightens it in proportion to the quantity purchased.

The budget multiplier has an interpretation

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A binding budget does not by itself imply $v_m > 0$. → A counterexample

Roy’s identity: recover demand from value

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The next proof needs no differentiable demand or differentiable direct utility.

Roy without differentiating demand

Let $x^* \in D(p, m)$. LNS gives $p \cdot x^* = m$.

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If $v_m > 0$, every optimiser satisfies the same Roy formula, so demand is unique.

→ Back

Check Roy in the Cobb–Douglas example

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Likewise $x_2 = (1-\alpha)m/p_2$. The demand agrees with the direct solution.

Roy example 2: substitutes and a corner

For $u = ax_1 + bx_2$,

$$v(p, m) = m \max\left\{\frac{a}{p_1}, \frac{b}{p_2}\right\}.$$

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NES

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If $a/p_1 > b/p_2$, then $v = am/p_1$, $v_m = a/p_1$, $v_{p_1} = -am/p_1^2$ and $v_{p_2} = 0$.

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Roy gives $x_1 = m/p_1$, $x_2 = 0$.

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At the tie Roy's differentiability assumption fails; demand is a whole segment.

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Can the same derivative ratio recover all three demand regimes?

One ratio, three regimes

Compute $q^* = -v_r/v_m$ in each region.

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Recover $a^* = m - rq^*$. At $m = 4 - r$ and at $r = 4$, adjacent values and first derivatives agree.

Roy example 4: a kink in direct utility

NES

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Direct utility has a kink, but indirect utility is differentiable. The direct proof of Roy still applies.

Demand survives a change of utility scale

If $\widehat{v} = f(v)$ with $f'(v) > 0$, then

$$\widehat{v}_{p_i} = f'(v) v_{p_i}, \qquad \widehat{v}_m = f'(v) v_m.$$

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A strictly increasing transformation can have a zero derivative at a point; then the differentiable ratio may be unavailable.

Checkpoint

Which assumptions guarantee existence? Which give uniqueness?

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What two differentiability/positivity conditions are needed to use Roy's formula?

Read next, and keep the core route clear

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Homework 1 covers Lectures 1–2 and is due on 14 November 2026.

Next block: two ways to ask the consumer question

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Utility maximisation: given p, m, how much utility can the consumer attain?

Expenditure minimisation: given $p, \bar{u}$, how little spending attains the target?

One fixes resources; the other fixes a level of the same preferences.

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e is the expenditure function; H is the Hicksian demand correspondence.

Assume the target is attainable. A singleton is denoted $h(p, \bar{u})$.

Why an unbounded feasible set can still have a minimum

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This set is nonempty, closed and bounded, hence compact.

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A continuous cost attains a minimum there; excluded bundles are more expensive. → Details

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No excess utility uses continuity and positive prices, not LNS.

The expenditure function has a different curvature

NES

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The expenditure function has a different curvature

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Homogeneity of degree one: $e(tp, \bar{u}) = te(p, \bar{u})$.

It is nondecreasing in prices and, above $u(0)$ on attainable targets, strictly increasing in the target.

The expenditure function has a different curvature

NES

Homogeneity of degree one: $e(tp, \bar{u}) = te(p, \bar{u})$.

It is nondecreasing in prices and, above $u(0)$ on attainable targets, strictly increasing in the target.

With continuous utility and LNS, expenditure is continuous on the positive-price, attainable-target domain.

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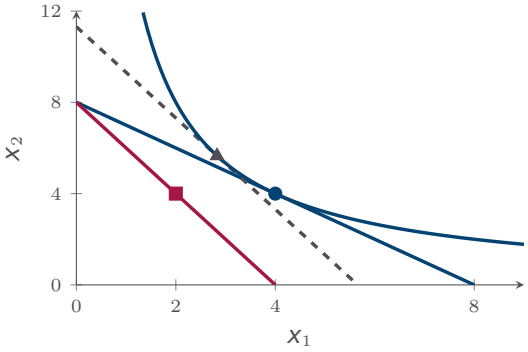
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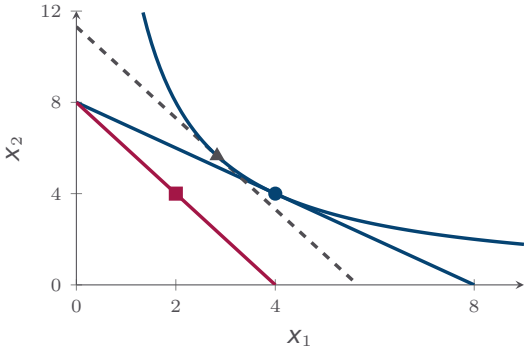
It is the infimum of linear cost functions. → Proofs

Fixed wealth versus fixed utility



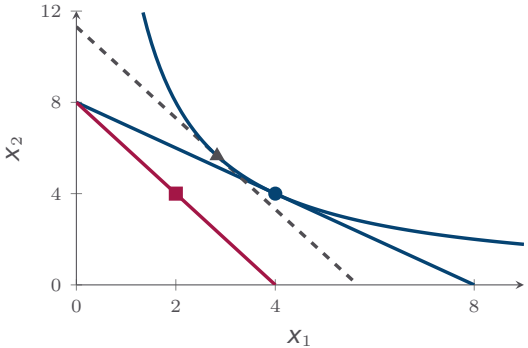
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Raise p_1 to 2: Marshallian demand at the same wealth is $(2, 4)$.
To keep utility 4, spend $8\sqrt{2}$ and buy $(2\sqrt{2}, 4\sqrt{2})$.

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These connect the two problems; they are not an assumption that the consumer literally solves both.

A monetary compensation question

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A money-metric change is useful, but summing it across people still involves welfare judgements.

Why the constraint qualification holds here

Active nonnegativity constraints have gradients e_i for the coordinates at zero.

→ Back to necessity and sufficiency

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The active gradients are therefore linearly independent. This includes corner optima.

→ Back to necessity and sufficiency

Proof: why concavity makes KKT sufficient

NES

For feasible y, concavity and differentiability imply

$$u(y) - u(x^*) \leq \nabla u(x^*) \cdot (y - x^*).$$

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Thus $u(y) \leq u(x^*)$ for every feasible y.

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Both saddle inequalities hold. → Back to the bridge

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The function is strictly increasing and quasiconcave. That alone does not make stationarity sufficient.

→ Back

A binding budget with zero marginal value

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Roy's ratio becomes $0/0$ at that point, although demand $x = m/p$ is perfectly well-defined.

→ Back to the multiplier

Cobb–Douglas with any positive exponent sum

Let $u(x) = \prod_{i=1}^{L} x_i^{a_i}$ with $a_i > 0$ and $A = \sum_i a_i$.

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No restriction $A \leq 1$ is needed for this preference solution.

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Proof: expenditure minimisation has a solution

Choose an attainable target and a feasible $\widehat{x}$. Let

$$S = \{x \geq 0 : u(x) \geq \bar{u}\} \cap \{x : p \cdot x \leq p \cdot \widehat{x}\}.$$

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Cost is continuous, so it has a minimum on S. Every omitted feasible bundle costs more than $\widehat{x}$, so this is a global minimum.

→ Back to the preview

Proof: homogeneity and price concavity of expenditure

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Scaling prices multiplies every feasible cost by the same positive number:

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$$e(p^t, \bar{u}) = tp \cdot x^t + (1-t)q \cdot x^t \geq te(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

→ Back to expenditure properties